

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Case Study

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Assessment Tool Weightage: 20%

Dave's Taxonomy: Precision (Level 3)
Question Count: 3

Total Marks: 30

Code of Conduct

I Beffy A Shanika(192511387) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

beffy

Assessment Tasks

Case Study

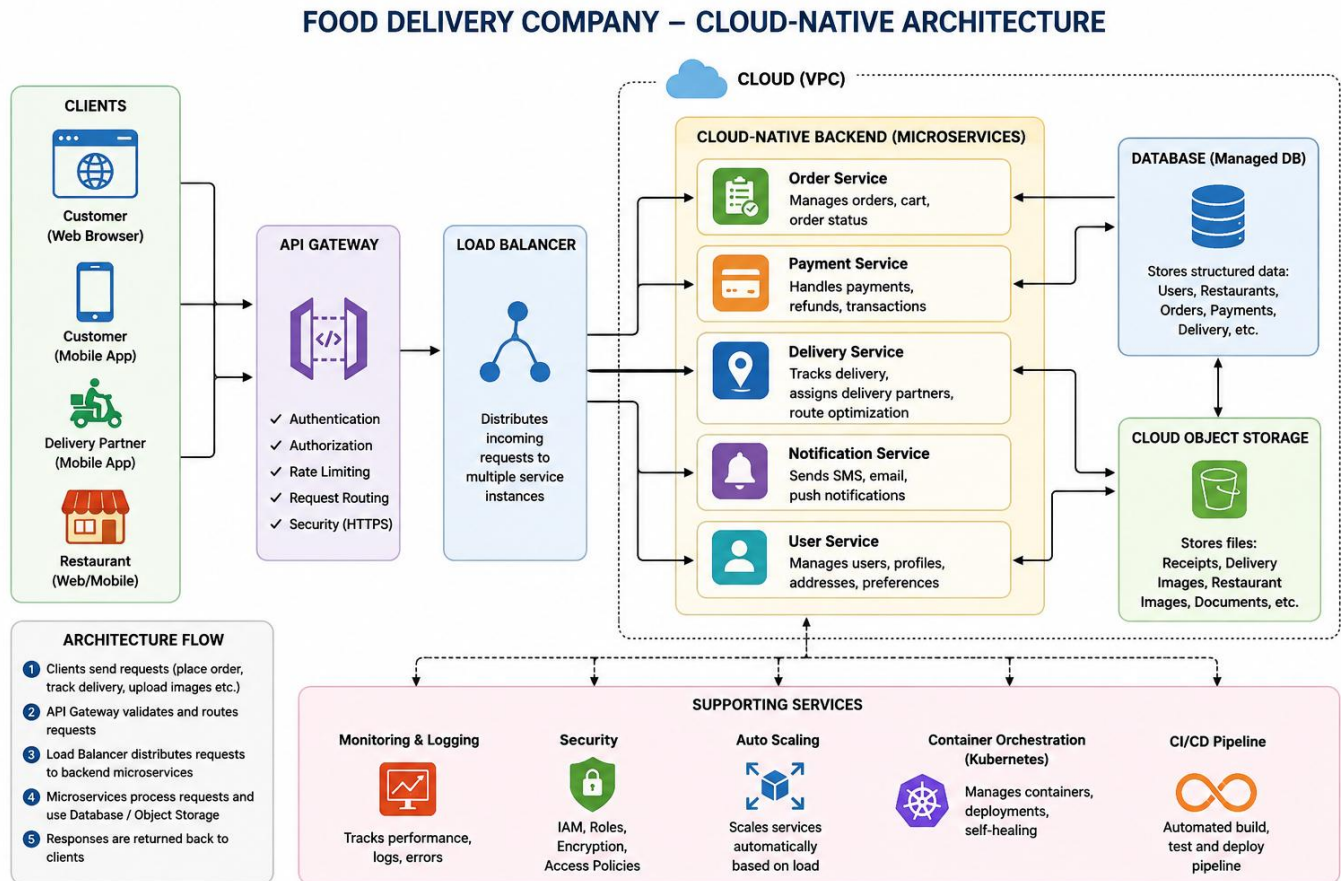
1. Case 1: A food delivery company wants to deploy a cloud-native backend with web APIs, browser clients, and shared cloud storage for receipts and delivery images. Propose the architecture and explain the role of each component.
2. Case 2: A media startup must decide between SAN-backed storage, object storage, and serverless processing for video upload workflows. Analyze the case and recommend a deployment model.
3. Case 3: A smart city platform needs low-latency decision making near sensors while still using the cloud for analytics. Analyze the use of edge computing and fog computing in this environment.

Case Study Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Case	25%	Clearly identifies all technical and business requirements	Identifies most requirements	Basic understanding only	Poor understanding of case
Application of Concepts	25%	Applies suitable cloud concepts and tools effectively	Good application with minor issues	Partial application	Weak or incorrect application
Analysis	20%	Strong analysis with clear trade-off reasoning	Reasonable analysis	Limited analysis	Minimal analysis
Solution Design	20%	Solution is feasible, practical, and well justified	Reasonable solution with minor gaps	Basic solution	Unclear solution
Clarity	10%	Well-structured and easy to follow	Mostly clear	Somewhat unclear	Poorly organized

Case Study 1: Food Delivery Company – Cloud-Native Backend

Proposed Architecture:



Proposed Architecture Components

A cloud-native architecture can be designed using the following components:

1. Browser/Mobile Clients

- Customers, delivery partners, and restaurants access the system through web or mobile applications.
- They send requests such as placing orders, tracking deliveries, and uploading images.

2. Web/API Gateway

- Acts as the entry point for client requests.
- Handles authentication, routing, rate limiting, and security.

3. Cloud-Native Backend

- Uses containerized microservices for order management, payment, delivery tracking, notifications, etc.
- Services can be independently scaled based on demand.

4. Database

- Stores customer details, restaurant information, orders, payments, and delivery status.
- A managed cloud database can reduce administration.

5. Cloud Object Storage

- Stores receipts, food images, delivery proof images, and other files.
- Provides scalable and durable storage.

6. Load Balancer

- Distributes incoming requests across multiple backend instances.
- Improves availability and prevents overload.

7. Monitoring and Security

- Monitors application performance and errors.
- Authentication, authorization, encryption, and access policies protect data.

Conclusion:

This architecture provides scalability, availability, security, and easy maintenance. Cloud object storage is suitable for large files, while microservices efficiently handle food-delivery operations.

Case Study 2: Media Startup – Video Upload and Processing

A media startup needs storage and processing for large video files. The three options can be analyzed as follows:

Option	Advantages	Limitations
SAN Storage	High performance, low latency, suitable for block-level operations	Expensive and requires more management
Object Storage	Highly scalable, durable, cost-effective, suitable for large media files	Not ideal for low-latency block-level workloads
Serverless Processing	Automatic scaling, pay-per-use, no server management	May have execution limits and cold-start delays

Recommended Deployment Model

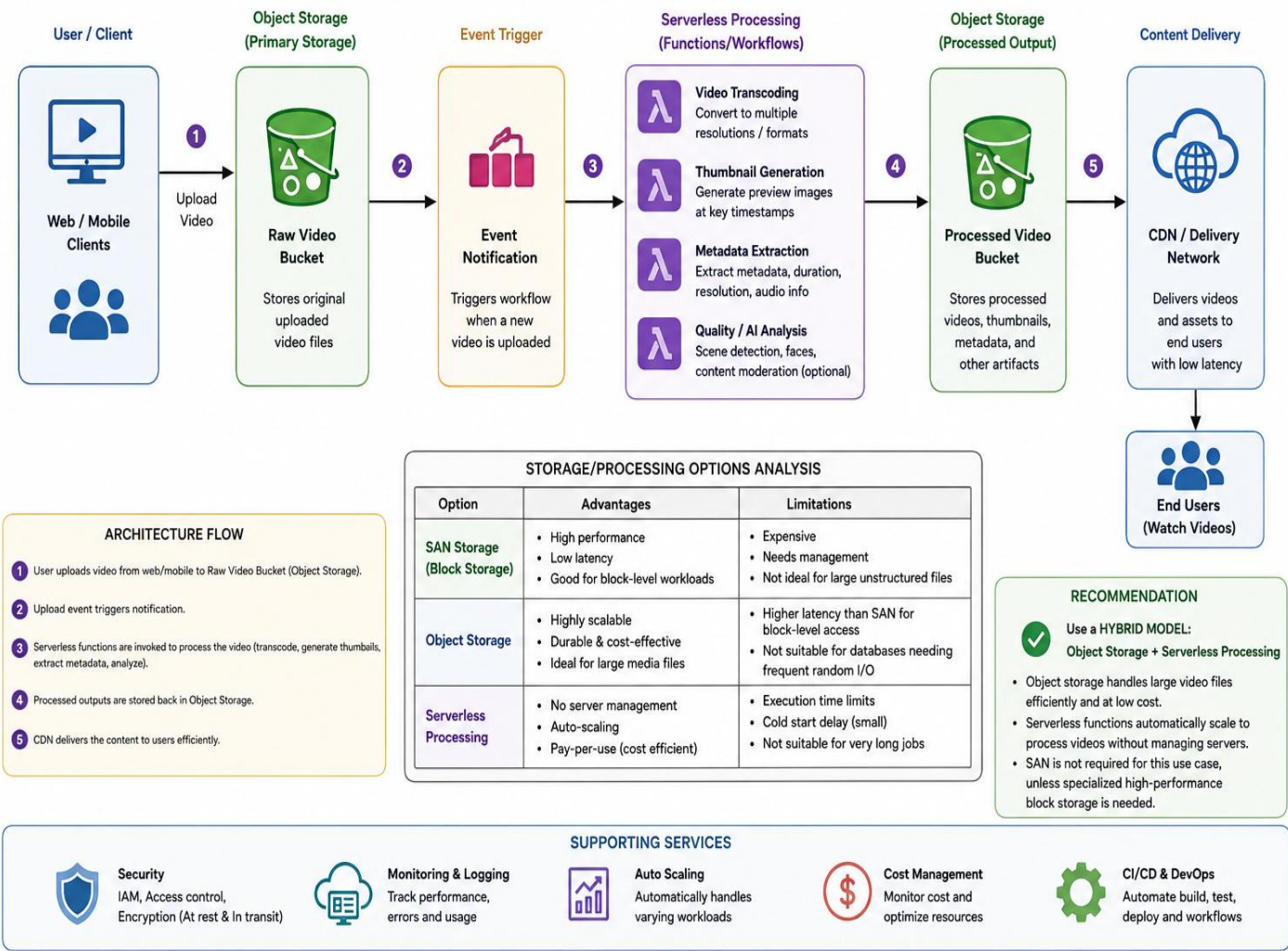
A hybrid object-storage + serverless model is recommended.

1. Users upload videos directly to cloud object storage.
2. Upload events automatically trigger serverless functions.
3. Serverless processing performs tasks such as:
 - Video format conversion
 - Thumbnail generation
 - Metadata extraction
 - Quality or resolution conversion
4. Processed videos are stored back in object storage.
5. A CDN can distribute videos efficiently to users.

Architecture Flow

Case Study 2 – Media Startup: Video Upload & Processing

Recommended Deployment Model: Object Storage + Serverless Processing



Conclusion

Object storage is the best primary storage because video files are large and require high scalability. Serverless processing avoids maintaining dedicated servers and automatically handles changing workloads. SAN can be used only when specialized high-performance block storage is required.

Case Study 3: Smart City – Edge and Fog Computing

A smart city may have thousands of sensors monitoring traffic, pollution, security cameras, water systems, and energy consumption. Sending every piece of data directly to the cloud can cause high latency and network traffic.

Edge Computing

Edge computing processes data close to the sensors or devices.

Examples:

- Traffic cameras detect accidents locally.
- Sensors identify abnormal pollution levels immediately.
- Smart traffic lights make decisions based on nearby traffic conditions.

Advantages:

- Very low latency
- Faster decision making
- Reduced bandwidth usage
- Can continue working even with temporary cloud connectivity problems

Fog Computing

Fog computing provides an intermediate computing layer between edge devices and the cloud.

For example:

Sensors → Edge Devices → Fog Nodes → Cloud

Fog nodes can collect and process data from multiple nearby sensors before sending important or summarized information to the cloud.

Role of Cloud Computing

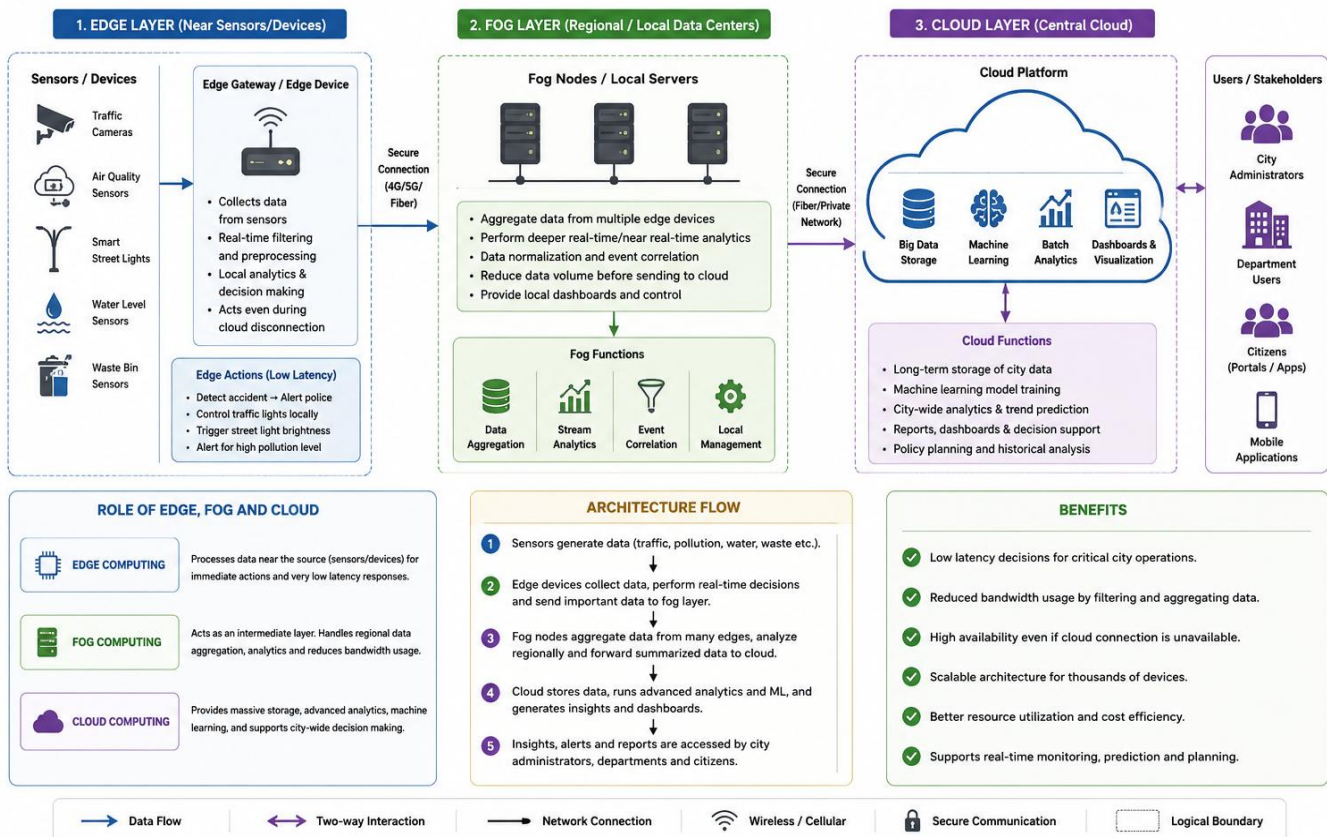
The cloud can be used for:

- Long-term data storage
- Large-scale analytics
- Machine learning
- City-wide monitoring
- Historical data analysis
- Generating reports and dashboards

Recommended Architecture

Case Study 3 – Smart City Platform: Edge, Fog and Cloud Architecture

Low-latency decisions near sensors with cloud for analytics and long-term storage



- **Edge:** Immediate, real-time decisions
- **Fog:** Regional aggregation and processing
- **Cloud:** Large-scale storage and analytics

Conclusion

A combination of edge, fog, and cloud computing is ideal for a smart city. Edge computing provides fast responses, fog computing reduces network load by processing regional data, and cloud computing performs large-scale analytics and long-term storage.